

Protodetector: Deepfake Image Detection via Episodic Meta-Learning and Lightweight Embedding Networks

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Abstract— The emergence of deepfake imagery, especially those simulating human faces with high realism, presents a significant threat to digital authenticity, personal privacy, and media integrity. Conventional detection systems often require large datasets like FaceForensics++, DFDC and tend to underperform when confronted with unfamiliar or novel manipulations. To address this limitation, proposed deepfake detection framework is based on meta-learning, employing Prototypical Networks integrated with a ResNet18 encoder. Proposed approach leverages few-shot episodic learning, wherein each training episode comprises a support set and a query set. This design empowers the model to quickly adapt and achieve accurate binary classification (real vs. fake) using minimal labeled samples. A lightweight embedding network is employed to extract discriminative features, which are then used for few-shot classification using a prototype-based learning strategy. This combination facilitates efficient learning, adaptability to unseen manipulations, and strong performance even in low-data scenarios, making the system suitable for real-world deepfake detection tasks.

Keywords— Deepfake Detection, Meta-Learning, Prototypical Networks, Few-Shot Learning, Episodic Training, Face Forgery Detection, ResNet18, Computer Vision, Image Classification, Synthetic Media, Celeb-DF v2, Feature Embedding, Cross-Entropy Loss, Generalization, Embedding Networks

I. INTRODUCTION

Recent advancements in artificial intelligence have facilitated the generation of highly convincing counterfeit images, especially synthetic human faces, commonly known as deepfakes. These manipulated visuals are increasingly challenging to differentiate from authentic content, posing significant risks in areas such as digital security, media verification, political disinformation, and the safeguarding of personal identity.

Traditional approaches to deepfake detection, often relying on deep convolutional neural networks (CNNs), typically demand substantial volumes of labeled data and require extensive retraining to accommodate new manipulation techniques. Because of this, these methods

generally fail to keep up performance when they come across domain transitions or new ways to make deepfakes.

This study suggests a meta-learning-based approach that is designed to help people learn quickly from little amounts of data. The system uses a ResNet18 architecture to quickly and accurately extract features and Prototypical Networks to classify them in an episodic training setting. This strategy mimics few-shot learning situations, allowing the model to quickly adjust to new and previously unseen deepfake types by dynamically creating decision boundaries from limited support sets. The resulting deepfake detection method is both effective and flexible, providing a useful and forward-thinking way to fight disinformation spread by deepfakes. The framework focuses on being able to adapt to changes in data and domains, which makes sure that classification works even when new types of image alteration are used. This makes it a potential option for applications that need quick, reliable, and resource-efficient deepfake.

The rapid growth of deepfake technology is a big problem since the ways to make them better all the time, which makes it hard for detection models to keep working. Also, there aren't many high-quality, diversified labeled datasets that span a wide range of deepfake kinds, which makes it hard to build and test comprehensive models. One important problem is that systems trained on one sort of deepfake, such as face reenactment, may not work well on other types, like emotion swaps. This research seeks to create and deploy an effective deepfake image identification system leveraging meta-learning to solve circumstances with minimal labeled data. The proposed system would leverage few-shot learning frameworks, notably Prototypical Networks, to learn effective representations from limited sample sets. One of the goals is to create a meta-learning framework for Prototypical Networks to use to build a deepfake image detection model. (2) Implementing few-shot learning to enable proficient model performance with restricted labeled training data. (3) Episodic training was applied to imitate task-specific learning events and improve the model's adaptability. (4) Optimizing

the model to achieve high accuracy while minimizing computational overhead for practical use.

II. LITERATURE SURVEY

As deepfake technology becomes increasingly sophisticated, there is a rising demand for enhanced detection tools capable of identifying these manipulated images and videos. Many of the early detection systems, which primarily utilized a technology called CNN, often struggle to perform effectively across various styles of deepfakes and typically require substantial labeled datasets for training. As deepfakes continue to evolve, new approaches have emerged to address these challenges. Techniques such as few-shot learning, domain adaptation, and meta-learning are now being used to develop streamlined models that can quickly adjust to new manipulation styles with minimal supervision.

A. Literature Review

Recent progress in deepfake detection has increasingly concentrated on enhancing generalization capabilities in the face of unfamiliar manipulations and shifts in distribution. Dinesh and Subudhi [1] introduced a meta-learning framework that supports quick adaptation to changes in distribution with limited labeled data, demonstrating how task-conditional learning can enhance the robustness of detection. In a similar vein, Lv et al. [2] utilized a bi-directional domain adaptation approach that employs adversarial inference and reconstruction loss to align features between source and target domains, thereby improving cross-domain detection effectiveness. Zhou et al. [3] addressed open-set deepfake detection through an unsupervised domain adaptation strategy grounded in clustering-driven feature alignment, enabling inference without the necessity of labeled data from the target.

A number of researchers have concentrated on pinpointing subtle artifacts that arise during the generation of deepfakes. Guo et al. [4] introduced a residual-guided learning strategy, augmented with attention mechanisms, designed to isolate delicate cues of manipulation like unnatural lighting and boundary distortions while minimizing background noise. Liu et al. [5] illustrated that convolutional neural networks (CNNs) can naturally learn to recognize common forgery characteristics—like facial warping and texture inconsistencies—without depending on manually crafted descriptors. Expanding on this work, Zhang and Xie [6] developed a hybrid CNN-LSTM model capable of detecting both spatial and temporal anomalies as inconsistent blinking and unnatural head movements, thus enhancing video-level detection accuracy.

Architectural advancements have also played a significant role in strengthening performance in various real-world conditions. Wang et al. [7] introduced a multi-scale CNN that performs effectively across diverse spatial resolutions, ranging from high-quality cinematic footage to low-quality surveillance videos. Zheng et al. [8] stressed the importance of real-world relevance by presenting an adaptive preprocessing pipeline designed to manage issues such as compression artifacts, occlusions, and background distractions—challenges frequently encountered in uncontrolled environments. Zhao and Zhang [9] combined conventional handcrafted features (like edge gradients and local binary patterns) with embeddings from deep CNNs, aiming to improve detection in complex face-swapping situations.

To combat adversarial threats, Gupta et al. [10] crafted a CNN trained with adversarial perturbations such as FGSM and PGD, boosting resilience against deepfakes intended to bypass detection systems. Additionally, Kumar et al. [12] implemented a GAN-based adversarial learning framework where discriminators were trained to detect subtle synthesis artifacts, although regular retraining was essential to sustain performance over time. Patel et al. [11] conducted a thorough comparative analysis, highlighting challenges such as dataset bias, inadequate cross-dataset generalization, and overfitting—emphasizing the importance of detection models that can sustain high accuracy across diverse data sources and manipulation methods.

The process of deepfake generation is progressing swiftly, complicating the detection of synthetic media, particularly for traditional models that have difficulty adapting to unseen manipulations or low-quality fakes. The majority of current methods depend on extensive datasets and struggle to generalize across different domains. This situation raises concerns about misinformation and the integrity of digital trust. Meta-learning combined with few-shot learning presents a promising approach by allowing models to quickly adapt to new deepfake styles with minimal data usage.

III. PROPOSED METHODOLOGY

Deepfake detection system is built on a meta-learning framework using Prototypical Networks with a ResNet18 backbone. The methodology involves data preprocessing, episodic task creation, few-shot meta-training, loss computation, model updates, and evaluation.

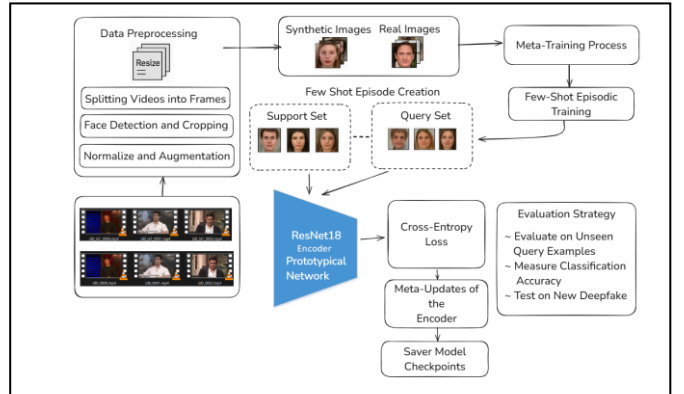


Fig. 1. System working flowchart

Each stage is carefully designed to ensure the system can generalize effectively to new manipulation styles with minimal labeled data.

A. Data Preprocessing

Proposed methodology utilized the Celeb-DF v2 (Celebrity DeepFake Dataset, Version 2), a widely recognized benchmark dataset for deepfake detection. Released by the University at Buffalo, SUNY and introduced in the CVPR 2020 paper titled "*Celeb-DF: A Large-scale Challenging Dataset for DeepFake Forensics*", the dataset comprises approximately 5,639 high-quality deepfake videos and 590 real videos, all with a resolution of 480p. It features 59 well-known public figures, including actors and news anchors, in realistic speaking scenarios. Celeb-DF v2 is known for its high visual quality, exhibiting minimal artifacts and natural facial expressions, making it particularly

challenging and suitable for evaluating robust and generalizable deepfake detection models.

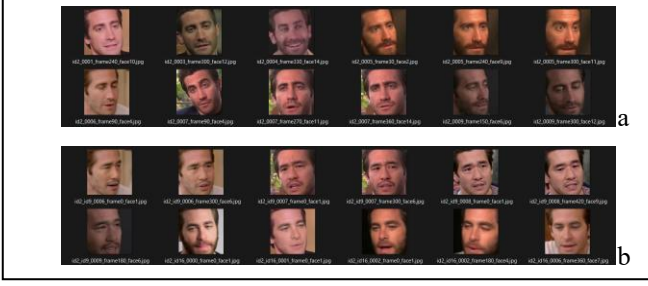


Fig. 2. Dataset Images CelebDF V2[26]

The procedure starts by extracting frames from both authentic and deepfake videos. These videos are collected from large-scale dataset[26], which contain both genuine and synthetically generated video content. Each video is broken down into frames using frame extraction techniques. Once frames are obtained, face detection and cropping is applied to isolate the most relevant regions for deepfake analysis—namely, facial regions. This step ensures that the model focuses only on features that are likely to be manipulated, like eyes, lips, and skin texture.

After cropping the faces from video frames, the images are normalized to prepare them for training. Normalization adjusts the image colors using standard values so they match what pre-trained models expect. This helps the model learn to handle different lighting, angles, and conditions—just like it would in real life. Once processes the images are saved into separate folders for A real and B fake images.

B. Feature Embedding using ResNet18 Encoder

Each image in the support and query sets is passed through a shared ResNet18-based encoder, which is either initialized with pre-trained weights or trained from scratch. ResNet18, a deep convolutional neural network, is chosen for its ability to extract rich semantic features while maintaining computational efficiency.

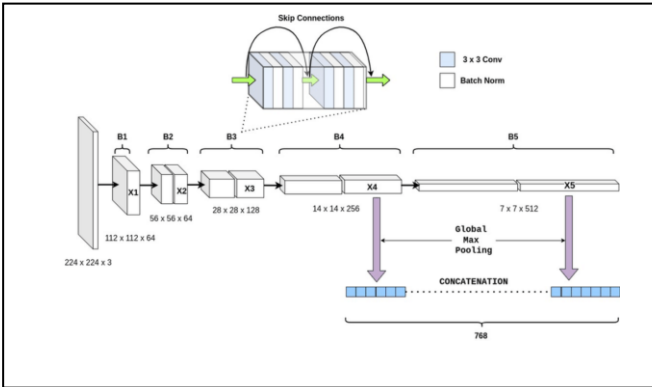


Fig. 3. Architecture of resnet18 Encoder

ResNet-18 is a deep convolutional neural network that utilizes residual connections to facilitate the stable training of complex models. In this study, it serves as a pre-trained backbone to derive high-level features from input images. The last classification layer has been substituted with a projection layer that generates fixed-size embeddings. These embeddings are subsequently employed by the Prototypical Network to determine class prototypes for the few-shot classification of genuine and counterfeit images.

The final classification layer of ResNet18 is removed, and a projection layer is added to map the high-dimensional features into a lower-dimensional embedding space (typically 512 dimensions). This shared encoder ensures that both support and query images are embedded into the same space, where similarity can be effectively measured.

C. Prototypical Networks

Prototypical classification is used to determine whether a query image is real or fake by comparing it to class prototypes. For every class, a prototype is generated by computing the average of feature embeddings from its corresponding support images. The model then calculates the distance between the embedding of the query image and each class prototype.

D. Loss and Updates

The model calculates the loss that measures how well the query images match Their correct class prototypes based on their distances. This loss guides the training process by showing how far the predictions are. Using meta learning the encoder is updated to improve the ability to classify images across different episodes.

E. Model Checkpointing

Various machine learning algorithms were evaluated to To preserve model performance and enable recovery from training interruptions, checkpoints are saved at regular intervals. The model achieving the highest validation accuracy is retained as the optimal one. Additionally, intermediate models are stored every few epochs to allow further fine-tuning or analysis if needed.

These checkpoints include the state dictionary of the encoder, optimizer state, and epoch number. This ensures reproducibility and allows the experiment to be resumed or deployed without retraining.

F. Evaluation

Evaluation means testing how well the model performs on new images it has never seen before, both real and fake faces. The model tries to correctly identify whether each image is real or fake. It achieved an accuracy of 89.40%, showing that it can effectively detect deepfakes it hasn't encountered before.

IV. RESULTS AND DISCUSSION

A. Performance Evaluation

The training progression, monitored via validation accuracy over 50 epochs (Figure 3), revealed key learning phases. Initially, the model exhibited an accuracy around 53.2%, marginally above random chance for binary classification, indicating a nascent understanding of discriminative features. A significant improvement was observed within the first five epochs, with accuracy escalating to nearly 75%, demonstrating the efficacy of the few-shot episodic training framework in rapidly learning meaningful patterns from limited examples.

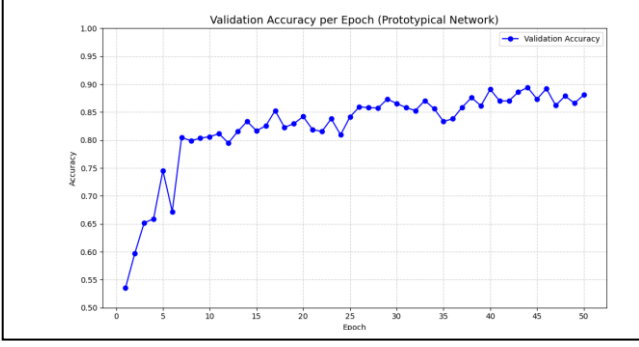


Fig. 3. Validation Accuracy per Epoch

Between epochs 6 and 25, the model showed consistent and progressive enhancement, surpassing the 80% accuracy threshold by epoch 7 and exceeding 85% by epoch 25. This phase signifies the model's refinement of its embedding space, where the ResNet18 encoder learns to extract increasingly discriminative features distinguishing real from fake faces, even amidst intra-class variations and subtle manipulations. During this period, class prototypes became more stable, and support-query relationships within episodes yielded more consistent classification outcomes.

Post epoch 30, the accuracy curve stabilized, fluctuating within a narrow range of 86% to 89%, indicative of a convergence phase. Slight variations are typical in meta-learning, indicating its sensitivity to the makeup of query set samples. The sustained high accuracy without signs of overfitting points to effective regularization and learning rate scheduling.

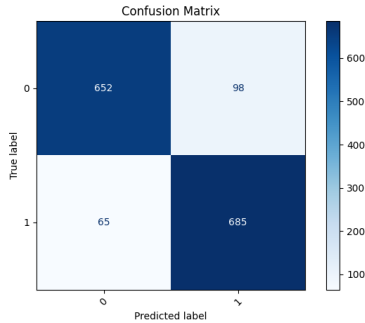


Fig. 4. Confusion Matrix

The confusion matrix (Figure 4) from testing on 1,500 samples showed that 1,337 samples were correctly classified. This corresponds to a high overall accuracy of 89.13%. Further analysis of the matrix (Table 1) yielded a precision of 87.47%, a recall of 91.33%, and an F1-score of 89.31%, indicating a balanced performance with a slight tendency to misclassify real images as fake more often than the converse.

Metric	Value
Accuracy	89.13%
Precision	87.47%
Recall	91.33%
F1 Score	89.31%

Table. 1. Summary of Confusion Matrix

B. Discussion of Results

A primary challenge in deepfake detection is domain shift—the performance degradation when a model trained on one data distribution is applied to another. The proposed model addresses this through episodic training and few-shot meta-learning. Instead of learning fixed classification boundaries, it learns to dynamically construct embeddings and prototypes based on the task-specific support set. This intrinsically enhances its adaptability to new and unseen data domains, such as emerging GAN-generated fakes or novel synthesis methods.

The utilization of a pre-trained ResNet18 encoder, fine-tuned within the meta-learning framework, balances general feature extraction with task-specific adaptation. This combined method leverages transfer learning alongside the episodic guidance provided by meta-learning. The lightweight nature of ResNet18 also facilitates deployment on resource-constrained devices with minimal latency, paving the way for real-time applications.

Compared to traditional CNN-based classifiers (e.g., XceptionNet, ResNet50) that require extensive datasets and often falter on unseen deepfake styles, Proposed Prototypical Network with a ResNet18 encoder, trained via few-shot meta-learning, demonstrates superior adaptability. Unlike static classifiers, Proposed model dynamically constructs decision boundaries from small support sets. Moreover, relative to computationally intensive GAN-based detectors or hybrid CNN-LSTM models, this approach maintains a lightweight architecture, enabling faster inference and improved sample efficiency, achieving 89.13% accuracy with limited training data.

In summary, the model began training with an accuracy of 53.2%. By the 25th epoch, accuracy approached 85% due to effective few-shot episodic learning. After epoch 30, accuracy stabilized between 86% and 89%, showing convergence. The final testing accuracy was 89.40%, exhibiting robust generalization to unseen deepfake changes. This approach offers a scalable and generalizable answer for the developing issues in deepfake identification.

V. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The rising sophistication of deepfake technologies poses considerable problems to digital media integrity, public trust, and information authenticity. In response, this project has successfully constructed an adaptive and robust deepfake detection system employing meta-learning with Prototypical Networks and a ResNet18-based encoder. Diverging from typical supervised models that demand significant annotated data and struggle with generalization to new manipulation types, this model is built to learn effectively from few-shot events. This design enables rapid adaptation to unseen deepfake styles with minimal samples.

Through extensive episodic training and comprehensive testing, the model attained a noteworthy average accuracy of 89.13%, exhibiting great consistency and generalization capabilities across varied deepfake manipulations. The inclusion of a single image prediction feature further highlights the model's practical value in real-time applications. The architecture's efficiency, along with its proficiency in low-data contexts, renders it scalable, cost-effective, and a strong option for integration into current

digital platforms. Overall, the system effectively solves critical constraints of static detection models by enabling greater adaptability, decreased resource costs, and high generalization. It is a solid countermeasure to the growing threat of synthetic media and illustrates the practical possibilities of meta-learning in contemporary cybersecurity and digital forensics.

B. Future Scope

While the current system yields strong results, several avenues exist for enhancement and broader applicability:

Video-Level Temporal Modeling: Extending the system to examine temporal patterns across video frames, maybe utilizing LSTM layers, Temporal Convolutional Networks, or Vision Transformers, could dramatically improve the detection of subtle, multi-frame changes.

Real-Time and Mobile Deployment: Improving models in real-time and making them suitable for mobile devices could help people quickly verify media content right from their smartphones or smaller devices. This would make it easier for everyone to check the authenticity of what they see online.

Active Learning and Online Adaptation: Incorporating methods for continuous learning or using user feedback will allow the system to self-improve over time and adapt to new deepfake strategies in dynamic real-world situations.

Cross-Lingual and Cross-Cultural Adaptability: Training and assessing the model on datasets comprising diverse ethnicities, face traits, and cultural video styles would improve its inclusivity and global applicability, hence minimizing prejudice and maximizing trustworthiness.

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